

Learned model — VALIDATION DOSSIER (does it work, and where is it sharper)

The living evidence file for `mirna_hallmark/learned/`. Every claim here is a **real number**, **a recovered positive control**, or **a specific sharpening/novelty example**, with the command to reproduce it. Companion to `LEARNED_MODEL_ESTIMATOR_MAP.md` (what estimator does which job) — this doc answers *does each one actually work, and what does it buy over the baseline*. Add to it as components are validated; keep the honest negatives in §4 (they are part of the evidence).

Status legend: [OK] verified this arc · [FILE] from a prior run (cited) · [TODO] to (re)generate under canonical M.

§1 Component gates — the summary numbers

Component	What it must show	Result	Reproduce
Adaptive-lasso coupling (the model)	learned $X \cdot M$ anti-correlates target OOF, FDR-controlled; beats raw abundance where weighting is applicable	[OK] genome-wide 1119 POOLED-HE genes + BH/BY-FDR (pooled-HE regen 2026-07-06): 597 (53%) FDR-significant coupling ($q_{BY} < 0.05$ & $\rho < 0$; q_{BH} 701). Learning beats abundance 356/597 among FDR-sig genes , 562/1119 (50%) overall — diluted by singletons where weighting is vacuous (§4). mean $\rho_{model} -0.123$. (<i>per-regulator-count breakdown, HE-era 35%→88%, to re-tabulate.</i>)	<code>mvp.gate_fdr → out-put/learned/gate_fdr.tsv</code>
Prior magnitude (does learning the magnitude matter, non-circularly)	functional evidence predicts repression more than binding-only, on external transfection data	[FILE] functional-evidence ρ +0.21 vs binding ρ +0.08 ($\sim 3\times$), within-family, McGeary GSE140217	<code>learned.evidence.calibrate</code>

Component	What it must show	Result	Reproduce
Prior robustness (is the result hostage to prior hyperparameters)	gate stable hand-set vs data-calibrated class weights	[FILE] PTEN -0.349 vs -0.348 ; gate 5/5 & 4/5; stability $0.85 \rightarrow 0.87$ — non-critical (prior used for <i>ordering</i>)	w_prior_source="ledger" vs "ledger_mrna"
Spike-and-slab inclusion (parked Bayesian alternative, now BUILT & benchmarked)	does an explicit evidence-graded inclusion prior $z \sim \text{Bernoulli}(\pi)$ beat the lasso's soft-L1 selection on OOF coupling?	[OK] it does NOT — head-to-head on the SAME folds/C/family predictors. Hub panel: beats abundance 4/5, recovers curated drivers at $\text{PIP} \approx 1.0$, but matches/beats lasso only 2/5 (mean ρ -0.313 vs lasso -0.351). Genome-wide sample (107 genes): FDR-sig 35% vs lasso 45% , mean ρ -0.078 vs -0.125 , matches/beats lasso on 21% . = inclusion prior buys a real per-edge PIP (interpretability) but not coupling; confirms the design decision to ship the lasso and park the spike-and-slab (rationale §2a)	learned.spike_slab (run()); gate_fdr(limit=120) $\rightarrow$ out- put/learned/gate_fdr_spike_slab.

Component	What it must show	Result	Reproduce
EB learned-shrinkage toward the prior (the τ^2 question)	make the curated prior a <i>location</i> and let the data learn the anchor strength τ^2 — does adaptive anchoring beat the fixed- α lasso?	[OK] coupling wash, but the learned τ^2 VALIDATES the design. $\beta_m \sim N(c \cdot \hat{w}_m, \tau^2)$, c/τ^2 learned by Gibbs. Hub panel mean ρ −0.344 vs lasso −0.351. Diagnostic τ^2/c^2 (data dispersion ÷ prior-anchored part) = 71-57000 in 4/5 genes = the data itself discounts the prior <i>magnitude</i> and shrinks little toward it (ESR1 the lone anchor-using gene, ratio 2.4, $c=0.27$). So “prior sets ordering not magnitude” is derived from data , not assumed. Side-finding: EB’s ZEB1 win (−0.596 vs −0.438) is Gaussian-ridge>L1 on a cooperative family, not the prior ($c \approx 0$).	<code>learned.eb_shrink.run()</code> (rationale §2c)

Component	What it must show	Result	Reproduce
Learned-τ^2 NN-ridge vs lasso — GENOME-WIDE (the coupling improvement that held)	does dense learned- τ^2 shrinkage beat the lasso's sparse selection on OOF coupling at scale, biology intact?	[OK] YES, significant. 655 multi-reg genes: ebridge_nn (L2, ν^2 learned, $M \geq 0$) mean ρ -0.1683 vs lasso -0.1518, wins 58% (379/655), Wilcoxon $p=9e-16$. nnridge_cv $\approx$ lasso $\Rightarrow$ it's the LEARNED τ^2 , not L2 (CV over-shrinks). Hyperprior-robust (InvGamma 0.01 $\rightarrow$ 20). CAVEATS: tail-driven (median Δ -0.003, mean -0.0165 — concentrated in many-regulator genes: CCL5, MET, E2F2, MMPs); collinearity hypothesis FAILS (Spearman(collin, Δ)=-0.074, non-monotonic) $\Rightarrow$ mechanism is regulator-COUNT not cooperative-families. Dense (no selection) $\Rightarrow$ better for the coupling job, not a lasso drop-in.	learned.shrinkage_compare.full() $\rightarrow$ out-put/learned/shrinkage_compare.ts (rationale §2d)
Learned-τ^2 for ATTRIBUTION (EB posterior vs bagged NNLS)	can the same τ^2 model read the per-edge weight AND give calibrated uncertainty, matching bagged NNLS?	[OK] hub: reproducibility EB 0.996 $\geq$ NNLS 0.983 (single-lasso 0.03); agreement ρ 0.844; identified set	mean/sd
Bagged-NNLS attribution (stable coefficients)	per-edge weight reproducible under resampling (single lasso is not)	[OK] single-fit bootstrap corr 0.03 $\rightarrow$ bagged 0.99 (PTEN); ESR1 0.93, ZEB1 0.99	states.canonical_M(g,"01") two seeds

Component	What it must show	Result	Reproduce
Family collapse + conditioned nomination	classify family as single-driver / co-drivers / family-only by conditioned anti-corr	[OK] §3.2: 13:7:11 (PTEN miR-17~92 = miR-106b+miR-20a co-drivers ; ESR1 miR-221/222 family-only)	families.resolution_report(g)
Identity vs magnitude (Decision I)	who EXPLAINS the target (Shapley R^2) vs who exerts force (budget $M \cdot \bar{X}$)	[OK] diverge by abundance-vs-coupling: ESR1 miR-18 explains 0.54 on 0.20 budget (quiet on-target owner, indiv $\rho -0.43$); miR-22 budget 0.26 / identity 0.01 (loud passenger, indiv $\rho -0.12$)	attribution.identity_vs_magnitude
CN-locus instrument (causality) + cluster-exclusion	miRNA-locus CN $\rightarrow$ abundance $\rightarrow$ target (reduced <0), AND the effect is attributable to the specific arm, not its genomic cluster	[OK] 60 edges: 59/60 usable ($F>10$), 46 causal-concordant , but after the cluster check only 24 STRONG (17 cluster-clean + 7 pleiotropic-but-arm-unique) . Clean-strong: miR-30d $\rightarrow$ GNAI2 (reduced -0.41), miR-200a $\rightarrow$ CTNNB1, let-7b $\rightarrow$ IFNB1; rescued: miR-96 $\rightarrow$ ADCY6/FOXO1 (co-targeter miR-182 flips +), miR-141 $\rightarrow$ PTEN (-0.28)	instrument.run_clean(60) $\rightarrow$ out- put/learned/instrument/cn_instrument
Cross-gene pooling (honest negative)	pooling should help weak/small-n genes IF genes share regulators	[OK] does NOT help on p53 set ($\Delta\rho -0.002$ full, -0.045 at $n=150$); neutral on miR-29 module ($+0.001$) — structure-bound, see §4	hierarchical.oof(); pro- gram_network.module_pooling_test

Component	What it must show	Result	Reproduce
Subtype-specificity	LEVEL vs common-M SLOPE vs WITHIN-subtype wiring (independent vs pooled)	[OK] pressure LEVEL differs 10/10 (Kruskal $q \approx 0$); common-M SLOPE differs 2/10 (RB1,CDKN1A). Independent two-fit subtype_wiring <i>suggested</i> PTEN wiring-remodels — but the pooled Bayesian-nesting estimator (subtype_wiring_pooled, H: ridge $M_s = M_{all} + \delta$, regularised Δ vs n-matched null) shows that was a noise-difference over-call: PTEN is borderline $\sim 1.35\times$ across $\lambda = 1-30$ (not robust); only RB1-Her2 modest ($\sim 1.5-1.66\times$). $\Rightarrow$ subtype wiring mostly CONSERVED; robust subtype signals = LEVEL + common-M slope, not rewiring	learned.subtype (subtype_coupling / subtype_wiring / sub-type_wiring_pooled)

§2 Positive controls — recovered known biology

The model recovers canonical, independently-established miRNA→target biology **without being told the answer** (the prior is inclusion+evidence, not the in-cohort magnitudes):

- **miR-21 → PTEN** — canonical oncomiR/tumour-suppressor axis. Recovered as PTEN's **rank-1 budget driver** (share 0.31) and a top learned weight. [OK] `states.budget_shift("PTEN")`
- **miR-200 / miR-429 → ZEB1** — the canonical EMT double-negative switch. Recovered as ZEB1's drivers, and the wiring is **conserved healthy→tumour** (transfer retention 0.93 — the switch is intact, only abundance shifts). [FILE] `states.stable_wiring("ZEB1")`, `state.cross_state_transfer`
- **miR-29 → collagen/ECM** — established ECM-repressor. Recovered as the **broadest EMT hub** (21 program targets, ADAM12/COL/LOX/MMP2). [OK] `program_network EMT`
- **miR-18 → ESR1, miR-34/449 → MET** — recovered as top drivers of their curated targets. [OK] §3

§3 Sharpening & novelty — specific examples

The vignettes that show the method is *sharper* than the abundance/evidence baselines, and finds new biology.

3.1 Learning re-weights off literature-prominence onto in-cohort signal (MET)

The **frozen evidence weights give the wrong sign** for MET (rho_curated **+0.23**, i.e. “more miRNA → more mRNA”) because they are dominated by the *most-studied* edges. Learning: - **keeps** miR-34/449 (evidence rank 1 — the one strong edge that also tracks MET), - **data-selects** miR-128 (rank 12) and **miR-181a-3p (rank 21 — near the bottom)**, - **drops** the well-studied-but-flat miR-1/206, miR-23, miR-199, miR-31, miR-144 (evidence ranks 2–6),

recovering correct repression (rho_model **−0.17**, gain +0.40). Same pattern on IGF1/ERBB2/E2F1/MYC. [OK] This is the core value: **literature prominence ≠ in-cohort predictive strength**, and the model corrects it.

3.2 Conditioned nomination resolves the driver(s) inside a collinear family — by distribution

Conditioning each family member on its family-mates classifies the family (§ METHODS 9) into **single-driver / co-drivers / family-only** by how many members’ anti-corr survives (part < −0.1). Across the panel: **13 single-driver : 7 co-drivers : 11 family-only**. Key cases: - **miR-17~92/106b → PTEN is CO-DRIVERS: miR-106b (−0.22) + miR-20a (−0.13)** — *not* miR-17 (flips to +0.09 once conditioned = it was riding the aggregate). miR-106b is the *strongest* survivor but not the sole one. (An earlier “miR-106b is *the* driver” oversimplified; the co-driver-aware nomination is the correct read.) - **ZEB1 = miR-429 + miR-200c** together (the canonical miR-200 family, co-drivers), miR-200b flips positive. - **family-only** (unidentified): ESR1 miR-221/222 (both ~−0.05, collinear) — the model honestly says so. The old abundance-divergence heuristic saw none of this (it never looked at the target). [OK] families.resolution_report → output/learned/attribution/resolution_panel.tsv (+ budget_shift_panel.tsv, 382 rows/20 genes)

3.3 The program network surfaces the regulatory architecture (all 5 Hallmarks)

Rolled up + persisted for all five priority programs (output/learned/program_network/). Each is a sparse network (density ~0.02) **concentrated on a few broad hub families holding 73–82% of program weight**:

Program	genes × families	edges	hubs (≥2)	hub weight	broadest hub (n_targets)
EMT	110 × 153	342	60	76%	miR-29 (21, ADAM12)
P53	88 × 152	328	70	82%	miR-27 (10, BTG2); miR-25/32/92 (9, TP53)
G2M	95 × 167	310	66	73%	let-7/98 (8, AURKB)
IFN	67 × 141	247	47	73%	miR-17~92 (12, ZNFX1)

Program	genes × families	edges	hubs (≥2)	hub weight	broadest hub (n_targets)
HYPOXIA	92 × 160	328	68	81%	miR-155 (9, MXI1)

The focus metric (max/sum) separates broad regulators (miR-29 = 0.13) from narrow ones (miR-21 = 0.68). **Positive control — the recovered hubs are the canonical program regulators:** miR-17~92/106 (cell-cycle /IFN), let-7/98 (G2M — AURKB), miR-155 (hypoxia), miR-29 & miR-200 (EMT), miR-15/16 (P53/G2M). And a cross-program readout the per-gene view can't give: the **miR-17~92/106 cluster and the miR-15/16/195/424/497 cluster are top hubs in 3 programs each** — genuinely pleiotropic program regulators. [OK] program_network EMT|P53|G2M|IFN|HYPOXIA

3.4 Discovery — validated novel edges (full genome-wide run, POOLED-HE 2026-07-06)

[OK] Full pass over **1571 pooled-HE genes: 2955 candidate orphan edges vs 2 permutation-null hits → edge-level FDR ≈ 0.001**; after composition control, **1579 deconv-robust across 460 genes, 289 fully novel**. (*Just below the HE-era 3034/1589/291 because the ~640 TarBase functional edges are now KNOWN, not orphan discoveries.*) - Top robust hits cross-validate the day's other findings: **miR-135b→GATA3** (partial_deconv -0.42, novel, retention 0.84) and the **miR-17~92/106b cluster → RABEP1, TGFBRAP1, AHNK, IL6ST, AFF1, M6PR, WWP1, SYNE1** — nominated by **miR-106b/93/17**, the same driver the 4b conditioning picked inside the collinear family. - **miR-135b→GATA3** also validated on CPTAC **protein -0.49** + literature ceRNA axis (spiker, 28%>floor = coupling in the miR-135b-high subset). [FILE] 111 orphans: 60% carry chimeric-eCLIP duplexes (Manakov). - Output: output/learned/discoveries.tsv (1579 robust), discovery_candidates.tsv (2955). discovery.run_all - **Program enrichment** (learned.enrichment, hypergeometric + BH): the 462 robust-target genes are FDR-enriched in **TGF_BETA_SIGNALING** (fold 1.7, q=0.043) and **UV_RESPONSE_DN** (q=0.028). Hub families (miR-106b/17) show **no single-program concentration** (nominal TNFα-NFκB p=0.015, q n.s.) — coherent with their cross-program pleiotropy (§3.3): broad regulators don't enrich one program. - **TGF-β composition-robustness:** the TGF-β edges survive the finer **myCAF + within-malignant-EMT** controls (assemble_gene(mycaf=True, mal_empt=True)) — median coupling retention **0.90** (XIAP/ACVR1/ SMURF1-2/ARID4B ≥0.9; only the EMT-linked KLF10 edges drop to ~0.65). Not a lumped-CAF or EMT-state artifact. - [TODO] **FDR is currently edge-level (flat)** — should be recomputed at the **seed-family** unit (Simes-per-family BH + per-family min-statistic null, mirroring coupling_inference.py) so family-mate orphans aren't double-counted. See §5.

§4 Honest negatives & caveats — also evidence

These bound the claims and show the gates are real (things fail them):

- **Cross-gene pooling doesn't earn its place (§1).** Fails on the p53/cell-cycle set (hurts at small-n), only neutral/marginal on the miR-29 ECM module — because the HE network is sparse (density 0.02) and effects are gene-specific. ⇒ per-gene is primary; program_network roll-up is the program view (aggregate *after* fitting, no pooling bias).
- **Occupancy saturating link failed the held-out gate** → linear a·M kept (occupancy.py parked); cooperativity/ceRNA gated behind it.

- **Prior hyperparameters are non-critical** — a feature, not a null: the result does not hinge on the class weights, because the model uses the prior for ordering/selection and estimates magnitudes from data.
- **Cross-state ΔM** compares an $n \approx 1000$ tumour fit to an $n = 104$ NAT fit — the small- n fit is more diffuse, so the wiring sum partly reflects n , not only biology (read the top movers, not the tail).
- **Composition confounding** — GTEx-breast (stroma-rich) vs tumour (epithelial) mismatch is a headwind against cross-state transfer; “conserved” (ZEB1) is the robust call, “remodeled” (PTEN/ESR1) is composition-suspect without deconvolved M_H .
- **“Beats abundance” is only meaningful for multi-regulator genes** — 396/1033 HE genes are singletons (one regulator family), where $X \cdot M$ has the same ranking as abundance up to a per-fold scalar (mean $|\Delta| = 0.03$, 47% exact ties); the OOF refit then only adds estimation variance (nothing to weight) $\rightarrow$ singletons “beat” just 19% (10% among FDR-sig). Not a model failure — the right tool for a singleton is the direct single-edge coupling (+FDR), not the aggregate. The model’s value is specifically multi-regulator genes: the equal-weight abundance baseline degrades as more (noisy) regulators are averaged in, so the weighting advantage grows with count (beat-rate $\rightarrow \sim 95\%$ at 8+). **Checked whether the singleton fit truly harms (2026-07-05): it does not** — mean harm $\rho_{\text{model}} - \rho_{\text{abund}} = +0.026$, and **0/104 real-coupling singletons lose their edge**; the harm is confined to the *gate metric*, while the attribution/card layer already reports singletons via direct partial-Spearman. So the penalty-bypass is not worth implementing (criterion: only if it really harms).
- **The CN instrument rejects some observationally-strong edges** — 14/60 are non-concordant; notably $\text{miR-17/20} \rightarrow \text{RUNX1}$ and $\rightarrow \text{CDKN1A}$ have strong *observational* coupling (-0.30 to -0.40) but ~ 0 reduced-form ($F > 60$, so not a weak-instrument artifact) — the causal test does not support them, a caution that observational anti-correlation $\neq$ causal repression for a subset of edges.
- **CN instrument is a CLUSTER perturbation** — of the 46 concordant edges, 22 are pleiotropic AND not uniquely attributable to the arm (a co-located co-targeter co-carries the effect), so the causal claim is cluster-level, not arm-specific. E.g. **miR-17 $\rightarrow$ PTEN** is concordant but miR-18a co-carries it (-0.24), and the **miR-200/429 $\rightarrow$ ZEB1** edges are irreducibly cluster-level (miR-429 always survives conditioning). The honest arm-specific causal set is the **24 “strong”** edges (§1), not all 46. The within-cluster attribution (does the arm uniquely keep its anti-corr conditioning on cluster-mates) is bounded-circular by design — CN supplies exogeneity, expression only resolves *which* cluster member carries it (`instrument.cluster_attribution`).

§5 Multiple-testing / FDR status (hypothesis families, precursor-style)

Principle (mirrors the precursor `coupling_inference.py`): the **seed family is the hypothesis unit** (Decision F — family $\rightarrow$ gene is identified, arm is a nomination), so family-mate arms are ONE test, not N. Controls used: **BH** (independence), **Benjamini-Yekutieli** (arbitrary dependence — what shared regulators induce), **Simes-per-family** (combine within-family $p \rightarrow$ BH across families), and a **per-family min-statistic permutation null** (family-wise error).

Test	hypothesis unit	FDR status
Gate (aggregate coupling)	gene	[TODO] genome-wide mvp.gate_fdr running — BH + BY across genes (BY for the shared-regulator dependence); family-collapse already removes arm redundancy <i>within</i> a gene
Discovery (orphan edges)	(family → gene)	[OK] edge-level permutation FDR ≈ 0.001; family collapse counted (3034→1878, robust→1100, novel→251). [TODO] a <i>proper</i> family-level FDR wants the per-family min-statistic scan null added to discovery.scan_all
Coupling (per-edge)	(family → gene)	[OK] learned.coupling — family-aware permutation partial-Spearman (Freedman-Lane, per-gene C incl. target_cn, seed-family dependence preserved) → p_perm / p_z / q_bh / q_by / q_family (Simes). permutation_pvalues confirmed generic (no precursor coupling); called per-gene for the per-gene target_cn. Validated (PTEN top = miR-106b −0.50). [OK] genome-wide (pooled-HE 2026-07-06): 5499 edges / 1507 genes → 1746 edge-sig (q_BY_global<0.05 & rho<0), 2222 family-sig → 1895 distinct significant seed-family→gene (output/learned/edge_coupling.tsv)
Instrument (causal)	edge / genomic cluster	[OK] reduced-form p + BY added (run_clean): 24 strong-causal → 18 also FDR-sig (q_BY<0.05); cluster-attribution in §1
Identifiability (hierarchical)	edge	β/sd >2 per edge, [TODO] no cross-edge FDR

Discovery is the one component with a real (permutation) null today; the rest need the p→BH/BY/Simes layer. gate_fdr is the first, reusing mirna_hallmark.stats.bh_fdr + coupling_inference.{benjamini_yekutieli,family_simes_fdr}.

Reproduce index

```
mvp.oof_gate(gene, w_prior_source="ledger", family=True)    # coupling gate (beats abundance / cur
mvp.gate_fdr()                                              # genome-
wide gate + BH/BY FDR (54% sig; 77% beat among multi-reg)
learned.coupling.run() / coupling.edge_coupling(gene)      # per-edge family-
aware permutation FDR (p_family)
learned.enrichment                                          # discovery/hub program enrichment (hypergeom+BH)
learned.subtype                                             # subtype-
specific coupling (Kruskal level + interaction slope)
instrument.run_clean(60)                                   # CN causal + cluster-
attribution (24 strong / 18 FDR-sig)
states.canonical_M(gene, "01")                             # canonical bagged attribution weight
attribution.identity_vs_magnitude(gene)                    # identity (Shapley coupling ownership) vs
states.budget_shift(gene)                                  # within-
gene budget, family-apportioned
card.gene_card(gene) / card.decompose(gene)                # per-
edge attribution card /  $\Delta$ M decomposition
families.resolution_report(gene)                           # conditioned driver nomination
program_network EMT|P53|G2M|IFN|HYPOXIA                   # program miRNA×gene network + summaries
program_network.module_pooling_test(anchor, genes)         # (b) pooling gate on a co-
regulated module
learned.evidence.calibrate                                 # transfection calibration of class weights
```